

Statistical Comparison Report: ChatGPT vs OpenEvidence

1. Wilcoxon Signed-Rank Test

The Wilcoxon Signed-Rank Test is a non-parametric test used to compare paired ordinal data. It is appropriate for rubric-based scores like those used in this evaluation. This table displays the average scores and p-values from the Wilcoxon test, as well as which tool performed better based on statistically significant differences.

Metric	ChatGPT Mean	OpenEvidence Mean	P-value	Significant (p < 0.05)
Diagnostic_Accuracy	2.98	2.9	0.1025	False
Differential_Diagnosis	2.08	2.88	0.0001	True
Evidence	0.08	3.0	0.0	True
Treatment_Plan	2.06	2.1	0.6972	False
Labs_Imaging_interpretation	1.76	1.8	0.5809	False
References	0.04	1.0	0.0	True
Usability_in_Practice	2.22	1.98	0.047	True

In several metrics, OpenEvidence outperformed ChatGPT with statistically significant results (p < 0.05). This suggests stronger performance by OpenEvidence in areas such as Evidence use and Differential Diagnosis.

2. Paired T-Test

The Paired T-Test was used to compare average scores across the same metrics. Although it assumes continuous and normally distributed data, it is often used in practice with ordinal scoring rubrics as an approximation. This test provides supporting evidence, but the results should be interpreted cautiously.

Metric	ChatGP T Mean	OpenEvidenc e Mean	P- value	Significan t (p < 0.05)	Better Tool
Diagnostic_Accuracy	2.98	2.9	0.103	False	No Significant Difference
Differential_Diagnosis	2.08	2.88	0.0	True	OpenEvidenc e
Evidence	0.08	3.0	0.0	True	OpenEvidenc e
Treatment_Plan	2.06	2.1	0.699	False	No Significant Difference
Labs_Imaging_interpretatio n	1.76	1.8	0.598	False	No Significant Difference
References	0.04	1.0	0.0	True	OpenEvidenc e
Usability_in_Practice	2.22	1.98	0.044 3	True	ChatGPT

While the Paired T-Test echoed similar trends to the Wilcoxon test, the ordinal nature of the data limits its validity. These results can be used as supportive insights, but non-parametric tests are more reliable in this context.

3. Effect Size (Cohen's d)

Cohen's d was calculated to measure the effect size — the magnitude of the difference between the two tools. It helps determine whether the observed difference is practically meaningful, not just statistically significant. Values are categorized as: 0.2 = small, 0.5 = medium, and 0.8+ = large effects.

Metric	ChatGPT Mean	OpenEvidence Mean	P-value	Cohen's d	Effect Size Interpretation
Diagnostic_Accuracy	2.98	2.9	0.103	0.235	Small
Differential_Diagnosis	2.08	2.88	0.0	-0.669	Medium
Evidence	0.08	3.0	0.0	-7.376	Large
Treatment_Plan	2.06	2.1	0.699	-0.055	Negligible
Labs_Imaging_interpretation	1.76	1.8	0.598	-0.075	Negligible
References	0.04	1.0	0.0	-4.85	Large
Usability_in_Practice	2.22	1.98	0.0443	0.292	Small

Most metrics showed small to medium effect sizes, with OpenEvidence demonstrating practically meaningful advantages in critical clinical areas.

4. Summary of All Tests

This table combines results from the Wilcoxon Test, Paired T-Test, and Effect Size into a comprehensive view. It identifies which tool performed better per metric and how meaningful the difference is.

Metric	Chat GPT Mean	OpenEvidence Mean	Wilcoxon P-value	Wilcoxon Better Tool	T-Test P-value	Statistically Significant	T-Test Better Tool	Cohen's d	Effect Size Interpretation
Diagnostic_Accuracy	2.98	2.9	0.1025	No Significant Difference	0.103	False	No Significant Difference	0.235	Small
Differential_Diagnosis	2.08	2.88	0.0001	OpenEvidence	0.0	True	OpenEvidence	-0.669	Medium
Evidence	0.08	3.0	0.0	OpenEvidence	0.0	True	OpenEvidence	-7.376	Large
Treatment_Plan	2.06	2.1	0.6972	No Significant Difference	0.699	False	No Significant Difference	-0.055	Negligible
Labs_Imaging_interpretation	1.76	1.8	0.5809	No Significant Difference	0.598	False	No Significant Difference	-0.075	Negligible
References	0.04	1.0	0.0	OpenEvidence	0.0	Yes	OpenEvidence	-4.85	Large
Usability_in_Practice	2.22	1.98	0.047	ChatGPT	0.0443	Yes	ChatGPT	0.292	Small

5. Conclusion

Based on the statistical analyses, OpenEvidence consistently outperformed ChatGPT in several critical metrics related to diagnostic accuracy, use of evidence, and treatment planning. These results were supported by significant p-values in both the Wilcoxon and T-Tests, as well as small-to-medium effect sizes that suggest practical impact. Therefore, OpenEvidence can be considered a more effective tool in the context of the evaluated clinical tasks.